

Based on the hackathon guidelines in

Build with AI_ Code for Communities — Second Edition.pdf and an inspection of your project setup, here is the breakdown of why your dataset loading failed, where and how to search and extract real public datasets, and how to integrate the cheapest Gemini model.

1. Why You Got the Dataset Loading Error & How to Fix It

Root Cause

Your project repository did not have its dependencies installed (node_modules was missing). When running `npm run data:load`, the command failed with:

'tsx' is not recognized as an internal or external command

Cannot find module '@electric-sql/pglite'

How to Fix and Load Datasets

Run the following commands in your project terminal:

bash

1. Install all dependencies (installs tsx, @electric-sql/pglite, drizzle, @google/genai, etc.)

`npm install`

2. (Optional) Fetch fresh data from public APIs (Open-Meteo, OpenStreetMap, NLEM)

`npm run data:fetch`

3. Load the data into the database (uses embedded PGLite WASM if PostgreSQL is not running)

`npm run data:load`

NOTE

If you don't have a local PostgreSQL/TimescaleDB container running, the script in

`scripts/load-data.ts` will automatically fall back to **embedded PGLite WASM** stored in `.pglite_data/`. You can also explicitly set `USE_PGLITE=true` in

`.env`.

2. What the Hackathon PDF Requires (Track 3)

The hackathon document defines **Track 03: Smart Health & Supply Chain Resilience:**

- **Theme:** Solving for India at Indian scale (Primary Health Centres / PHC network).
- **Core Task:** Real-time visibility into medicine stocks, bed availability, personnel attendance, demand forecasting, stock-out early warnings, and cross-district resource redistribution.
- **Public Data Requirement (Pages 7 & 9):**
 - "Real or realistic data — public datasets, sample data, or APIs where live data isn't available"
 - Specifically lists: **data.gov.in and Indian government open data portals, WHO health data, ISRO/Bhuvan satellite data, IMD (India Meteorological Department).**

3. Where to Search Real Datasets & How to Extract Public Data

Here are the authoritative portals matching the hackathon criteria and how to extract data for your Smart Health platform:

Data Type	Official Portal & URL	Exact Search Terms	How to Extract & Project Path
PHC & CHC Directory	data.gov.in (Open Government Data - OGD)	"All India Health Centres Directory", "Karnataka PHC list"	Download CSV/JSON with columns: State_Name, District_Name, Facility_Type, Latitude, Longitude. Place in data/manual/karnataka_health_centres_directory.csv. (<i>Automated mirror: scripts/fetch-facilities.ts queries OpenStreetMap Overpass API for healthcare facilities</i>).
Essential Medicines Catalog	cdsco.gov.in & mohfw.gov.in	"National List of Essential Medicines 2022", "NLEM 2022 PDF"	Download Gazette PDF; extract Primary Care tier medicines (Paracetamol, Amoxicillin, ORS, Albendazole, etc.). Already structured in data/processed/nlem_primary_care.json.
District Population & Catchment	censusindia.gov.in	"Primary Census Abstract 2011 Karnataka", "PCA District Data"	Download District Census Excel/CSV to extract district and rural population numbers to compute per-capita PHC medicine quotas. Kept in

Data Type	Official Portal & URL	Exact Search Terms	How to Extract & Project Path
			data/raw/population/karnataka_census_2011.json.
Disease Outbreak Surveillance	idsp.mohfw.gov.in (NCDC / IDSP)	"IDSP Weekly Outbreak Surveillance", "Karnataka disease weekly bulletin"	Extract seasonal disease surges (Monsoon Dengue/Malaria peak, Summer ADD/Cholera) to build demand forecasting curves. Mirrored in data/raw/seasonality/disease_seasonality_karnataka.json.
Weather & Climate Risks	mausam.imd.gov.in or open-meteo.com	"IMD Daily Rainfall Karnataka", Open-Meteo Historical Weather API	Automated by scripts/fetch-weather.ts using district coordinates to calculate precipitation, monsoon surges, and flood risks affecting PHC supply routes. Stored in data/raw/weather/karnataka_weather_daily.json.

4. How to Integrate Gemini API Key with the Cheapest Model

Your app is already equipped with @google/genai in

src/lib/services/gemini-service.ts.

Step 1: Get Your Free API Key

1. Go to [Google AI Studio](https://aistudio.google.com/).
2. Sign in with your Google Account and click **Create API Key**.
3. Copy the generated key.

Step 2: Choose the Model (Cost Comparison)

Model Name	Free Tier Available?	Pricing (Pay-as-you-go)	Best For
gemini-1.5-flash-8b	Yes (Free)	\$0.0375 / 1M input tokens \$0.15 / 1M output tokens	Cheapest model in the Gemini family. Ultra-fast for high-volume text parsing, inventory classification, and voice-to-JSON.
gemini-2.0-flash	Yes (Free)	\$0.10 / 1M input tokens \$0.40 / 1M output tokens	Recommended for Hackathons: Free tier gives 15 RPM / 1,500 requests/day at \$0.00 cost, with high-accuracy native function calling and vision.
gemini-1.5-flash	Yes (Free)	\$0.075 / 1M input tokens \$0.30 / 1M output tokens	Very stable, balanced, multimodal capabilities.

Step 3: Configure Your Environment File

Open your

.env file and configure:

env

Google Gemini API

GEMINI_API_KEY=AlzaSyYourActualKeyGoesHere

For the lowest-cost model:

GEMINI_MODEL=gemini-1.5-flash-8b

Or for the best free-tier multimodal/function calling:

GEMINI_MODEL=gemini-2.0-flash

Step 4: Verification

Your application uses this key across 4 features:

1. Gemini Copilot Agent (

src/app/copilot/page.tsx): Uses 5 native function tools

(get_forecast, get_alerts, get_redistribution_plan, get_phc_status, get_resilience_score).

2. Multimodal Shelf Photo Inspector (

src/app/api/copilot/vision/route.ts): Inspects medicine boxes via photo.

3. **Multilingual Voice Stock Ingestion (**

src/app/api/copilot/voice-parse/route.ts): Parses spoken English, Hindi, and Kannada into structured batch updates.

4. **Daily Executive Briefing (**

src/app/api/copilot/briefing/route.ts): Generates morning resilience briefings.

Once npm install finishes and you add your API key in

.env, run:

bash

npm run dev

and your application will start with live AI intelligence and seeded public health datasets.